

## Lab 5. Transformers and Attention

In this lab, we explore one of the most surprising and consequential features of many transformers – their ability to immediately incorporate new information “in-context” to perform unfamiliar tasks, given prompts. They accomplish this without changing any of the weights/biases in the network, seemingly solving a problem that had previously limited neural networks as models of human cognition since the 1950’s, making them faster learners, more useful aids, and more interesting interlocutors. Here, we hope to better understand a little about how this is actually working!

We use **GPT-2 XL** (a precursor to ChatGPT that is now openly available) to learn a simple **repetition rule** from examples in the prompt. Specifically our task:

- (1) cues the network with a specific one-token word (e.g., *dog*),
- (2) provides an arbitrary symbol (>>>) which we arbitrarily designate means "copy the preceding word exactly three times".
- (3) We present the proper completion of our repetition task, displaying the word 3 times.

For example:

**prompt: dog >>> dog dog dog**

**test: tree >>>....? [correct answer: tree tree tree].**

Notably, the transformer doesn’t get any feedback. It just sees some number of examples of the task completed successfully.

This is obviously kind of a silly task, but by analyzing how it works, we get some clues for how transformers may be able to integrate new information in a prompt successfully.

Instructions:

1. Choose an arbitrary task-cue (which we will call a “SEPARATOR”) which the network must learn means “copy the preceding word exactly three times”. In the example above, I used “<<<”. You can make up anything **as long as it is a single token**. The provided code will help you check this.

@@@

2. Test the model with your chosen cue. How many examples are necessary for this model to learn the copy task?

Response: I chose “@@@” as my arbitrary task-cue. Only one example was necessary for it to copy the word ‘moon’ three times. It also successfully generalized the pattern for the word ‘river.’

3. Can we identify which heads are likely responsible for this behavior? Use BertViz to visualize the attention patterns of heads at different layers. We specifically want to find heads that reliably look back at the token they need to copy when they receive the task-symbol.

Copy the image into your homework that provide evidence for:

- (a) one layer in which the heads **aren't** attending to the specific information that needs copied.

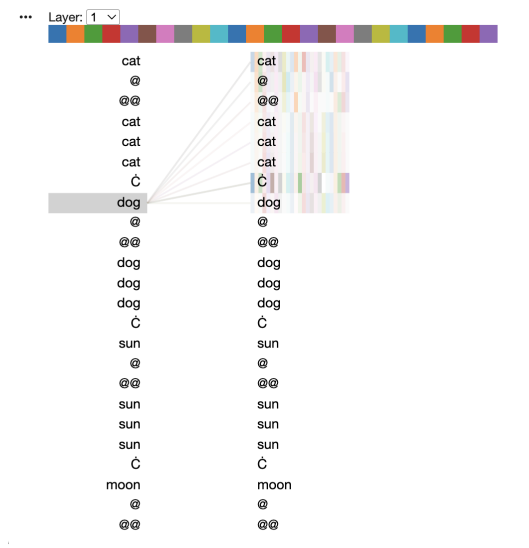


Figure 1. In layer 1, the word dog is falsely associated with cat. Attention is spread across many tokens instead of focusing on the specific information that needs copied.

- (b) one layer where they **are** attending to the specific information that needs to be copied.

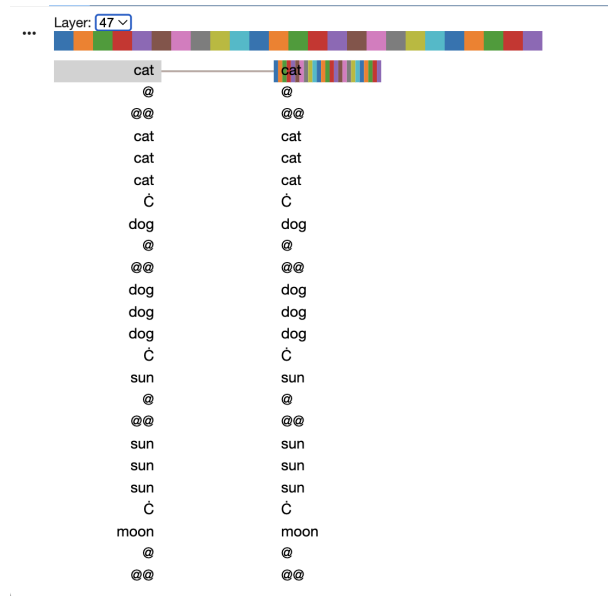


Figure 2. Layer 47 shows a higher level of attention to the token that needs to be copied. The first word ‘cat’ has only one, exact information on the right, which matches with the head. This demonstrates that a few examples are necessary for transformers to do well.

If I hover over the symbol, though, it is going to copy mostly the word ‘cat.’

- (c) Beyond looking at attention patterns, how could we provide stronger evidence for the role of particular heads in learning and using the new rule?

Response: I’m curious about the intersection of memory and attention. There are a lot of studies that look at patients with damage to their hippocampus, which often results in either a disruption of their memory or attention. If we remove the copying head, would transformers be able to do the task in a similarly accurate level as layer 47? Is it going to forget what has been copied just before (memory) or fail to copy the exact word three times (attention)? If it fails to copy the correct word, it would show that the head plays a major role in contributing to in-context learning.

*Bert Viz instructions: More instructions are provided in the colab as well as the in-class colab, but reminder: To use BertViz, first select a particular layer of the network using the drop down menu. Second, hover your cursor over a token to see the attention pattern for the heads (colors) in that layer. This gives us a sense of what those heads are attending to when they encounter that token. 3. Compare the attention patterns in an early layer (e.g., 1-5) to a later layer, such as 15. The most important heads **should attend specifically to the information they need to copy**).*

4. Conceptual Question: Throughout this course, we've seen three ways a system can store new information: (a) in its weights, through gradient descent over many examples (slow, permanent — like cortical consolidation); (b) in an external memory bank, through a

write head (fast, structured — like hippocampal binding); and (c) in the context window, through attention over the prompt (fast, ephemeral — gone when the conversation ends). In this lab, GPT-2 learned your novel task signal through method (c).

Provide

- (a)** one similarity between the GPT-2 mechanism and fast learning in humans and
- (b)** one difference.
  
- (c)** what could you potentially change about the model to mitigate the difference?

Response: (a) One similarity between the GPT-2 mechanism and the fast learning in humans is that both can extract new information to their approach without changing long-term memory. The GPT-2 learned the meaning of a '@' in a few examples and didn't have to update its weights. In humans, for example, we need to hear new information only so much to encode it into working memory.

(b) One difference between them is that GPT-2 loses the rule it has just learned because it's not rapidly updating long-term memory. Humans can apply new information quickly like we only need to hear someone's name once or twice to remember later. The GPT-2, however, is matching statistical patterns alone and limited to word-copying.

(c) We could add external memory to the transformer through a write head that stores new learned information beyond the current context window. Instead of forgetting them after the prompt ends, it would retain the information and would be able to reuse it for various contexts.